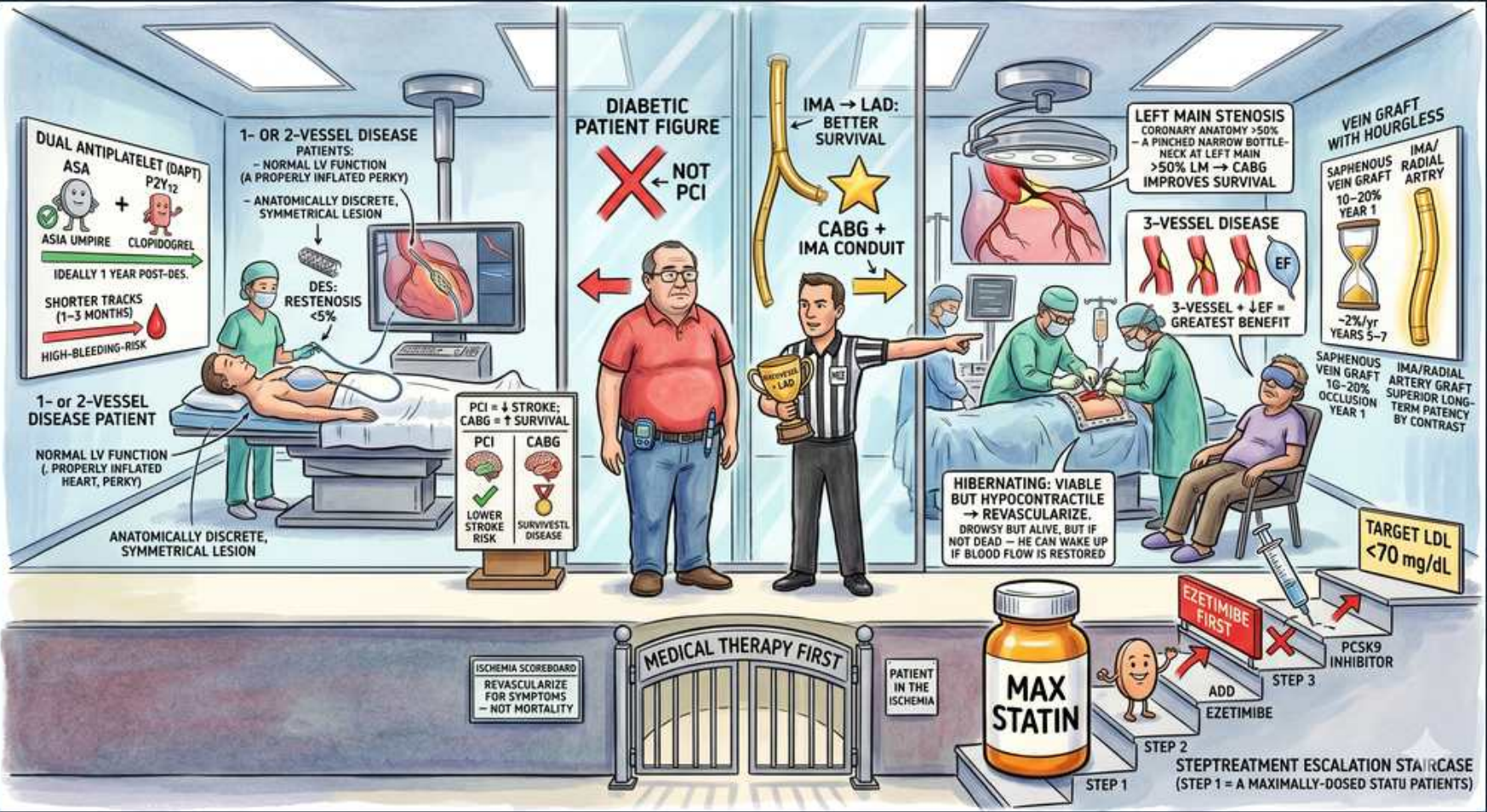


Revascularization in Stable IHD: CABG vs PCI & Medical Therapy



1. Diabetic patient → CABG, not PCI

WHAT YOU SEE

Diabetic figure, 'NOT PCI', CABG trophy

WHAT IT ENCODES

In diabetics with multivessel CAD, CABG is superior to PCI (FREEDOM trial) with lower death/MI. Diabetes drives diffuse, progressive disease that bypass grafts handle better than focal stenting. Board favorite: diabetic multivessel → CABG.

BOARD TRAP

PCI for a diabetic with multivessel disease — CABG is preferred; the discriminator is diabetes + multivessel anatomy.

2. 1- or 2-vessel disease → DAPT & PCI

WHAT YOU SEE

DAPT: ASA + P2Y12, DES, '1–2 vessel'

WHAT IT ENCODES

For 1–2 vessel disease with discrete, symmetrical lesions and normal LV function, PCI with drug-eluting stent plus dual antiplatelet therapy (aspirin + P2Y12) is appropriate. DAPT ideally 1 year post-DES.

BOARD TRAP

CABG for simple 1–2 vessel discrete disease — PCI is the less invasive, appropriate choice when anatomy is favorable.

3. DES restenosis vs stent thrombosis

WHAT YOU SEE

DES 'restenosis', shorter DAPT tradeoff

WHAT IT ENCODES

Drug-eluting stents reduce in-stent restenosis (neointimal hyperplasia, months later) versus bare-metal but require longer DAPT to prevent stent thrombosis (acute platelet thrombus, often fatal). Premature DAPT cessation → stent thrombosis.

BOARD TRAP

Confusing restenosis with stent thrombosis — restenosis is gradual re-narrowing; thrombosis is acute occlusion from stopping DAPT.

4. LIMA → LAD graft survival benefit

WHAT YOU SEE

IMA→LAD, 'better survival', CABG + IMA conduit

WHAT IT ENCODES

The internal mammary (IMA/LIMA) to LAD graft has >90% 10-year patency and confers a clear survival benefit; it is the cornerstone conduit of CABG. Arterial grafts vastly outlast saphenous vein grafts.

BOARD TRAP

Using only vein grafts in CABG — the LIMA-to-LAD arterial graft is what drives the survival advantage.

5. Left main stenosis → revascularize

WHAT YOU SEE

Left main anatomy, '>50%', supplies >75% LV', improves survival

WHAT IT ENCODES

Left main stenosis >50% supplies the majority of the LV myocardium; revascularization (CABG, or PCI in selected anatomy) improves survival. This is a survival-benefit indication independent of symptoms.

BOARD TRAP

Medical therapy alone for significant left main disease — left main is a hard revascularization indication with survival benefit.

6. 3-vessel disease → greatest CABG benefit

WHAT YOU SEE

3-VESSEL DISEASE, surgeons, 'greatest benefit'

WHAT IT ENCODES

Three-vessel disease (especially with reduced LV function or diabetes) derives the greatest survival benefit from CABG over PCI. Complex/diffuse multivessel anatomy (high SYNTAX score) favors surgery.

BOARD TRAP

Multivessel PCI in complex 3-vessel disease — high SYNTAX-score anatomy favors CABG for survival.

7. Vein graft failure timeline

WHAT YOU SEE

Saphenous vein graft, hourglasses ~1 yr / ~5 yr

WHAT IT ENCODES

Saphenous vein grafts fail in stages: early thrombosis (<1 month), intimal hyperplasia (months–1 year), and accelerated atherosclerosis (years). They lag far behind arterial grafts — basis for preferring the LIMA to LAD.

BOARD TRAP

Assuming vein grafts last as long as arterial grafts — SVGs degenerate over years; LIMA patency is far superior.

8. Hibernating myocardium → recovers

WHAT YOU SEE

Hibernating: 'viable but hypocontractile, ↓EF, recovers if revascularized'

WHAT IT ENCODES

Hibernating myocardium is viable but chronically hypoperfused and hypocontractile, depressing EF; contractile function recovers after revascularization once blood flow is restored. Viability testing (CMR/PET) identifies it.

BOARD TRAP

Calling low-EF segments irreversibly dead — hibernating (viable) myocardium regains function after revascularization.

9. PCI reduces strokes vs CABG

WHAT YOU SEE

PCI '↓ stroke vs CABG', lower periprocedural stroke

WHAT IT ENCODES

PCI carries a lower periprocedural stroke and shorter-recovery risk than CABG, while CABG offers better long-term survival in complex multivessel/diabetic disease. The PCI-vs-CABG choice balances stroke/recovery against durability.

BOARD TRAP

Ignoring stroke risk when both are options — PCI's lower periprocedural stroke favors it in stroke-prone patients with simple anatomy.

10. Medical therapy first / statin escalation

WHAT YOU SEE

MEDICAL THERAPY FIRST, max statin → ezetimibe → PCSK9, LDL <70

WHAT IT ENCODES

Optimal medical therapy is the foundation for stable IHD. Lipid lowering escalates stepwise: maximally tolerated high-intensity statin → add ezetimibe → add PCSK9 inhibitor, targeting LDL <70 mg/dL (<55 in very high risk).

BOARD TRAP

Jumping to PCSK9 inhibitor before maxing statin/ezetimibe — the LDL-lowering escalation order is statin → ezetimibe → PCSK9.
